DM5 du 7/10/2022

Ex1 : ⓐ $A(x) = \dfrac{1-\cos(2x)}{\sin(2x)} = \dfrac{1-(1-2\sin^2(x))}{2\sin(x)\cos(x)} = \dfrac{\sin(x)}{\cos(x)} = \tan(x)$

ⓑ $B(x) = \dfrac{\cos(x)-\sin(x)}{1-\tan(x)} = \dfrac{\cos(x)-\sin(x)}{1-\dfrac{\sin(x)}{\cos(x)}} = \cos(x)\dfrac{\cos(x)-\sin(x)}{\cos(x)-\sin(x)}$

$= \cos(x)$

Ex2 : ⓐ $f(t) = 2\cos(3t) - 2\sin(3t)$

$\rho = \sqrt{2^2+(-2)^2} = \sqrt{8} = 2\sqrt{2}$ $\quad \begin{cases} \cos\varphi = \dfrac{2}{2\sqrt{2}} = \dfrac{\sqrt{2}}{2} \\ \sin\varphi = \dfrac{-(-2)}{2\sqrt{2}} = \dfrac{\sqrt{2}}{2} \end{cases} \Rightarrow \varphi = \dfrac{\pi}{4}$

Donc $f(t) = 2\sqrt{2}\cos\left(3t+\dfrac{\pi}{4}\right)$

ⓑ $g(t) = -3\cos(\pi t) + \sqrt{3}\sin(\pi t) = 2\sqrt{3}\left(-\dfrac{\sqrt{3}}{2}\cos(\pi t) + \dfrac{1}{2}\sin(\pi t)\right)$

$= 2\sqrt{3}\left(\cos\left(\dfrac{-5\pi}{6}\right)\cos(\pi t) - \sin\left(\dfrac{-5\pi}{6}\right)\sin(\pi t)\right)$

$= 2\sqrt{3}\cos\left(\pi t - \dfrac{5\pi}{6}\right)$

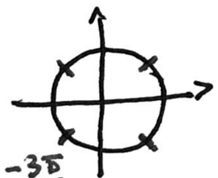
ⓒ $h(t) = \cancel{6}\sqrt{3}\cos\left(2t - \dfrac{\pi}{6}\right)$

Ex3 : ⓐ

$\cos(2t-\pi) = 0 \iff \{ 2t - \pi = \dfrac{-\pi}{2} + k\pi,\ k \in \mathbb{Z}$

$\iff 2t = \dfrac{\pi}{2} + k\pi\ ; k \in \mathbb{Z}$

$\iff t = \dfrac{\pi}{4} + k\dfrac{\pi}{2}\ ; k \in \mathbb{Z}$



$S = \left\{ \dfrac{\pi}{4} + k\dfrac{\pi}{2}\ ;\ k \in \mathbb{Z} \right\}$

ⓑ $2\sin\left(3t-\dfrac{\pi}{2}\right) - 1 = 0 \iff \sin\left(3t - \dfrac{\pi}{2}\right) = \dfrac{1}{2}$

(1) $\iff \begin{cases} 3t - \dfrac{\pi}{2} = \dfrac{\pi}{6} + 2k\pi \\ 3t - \dfrac{\pi}{2} = \dfrac{5\pi}{6} + 2k\pi \end{cases}$ $k \in \mathbb{Z}$

$\iff \begin{cases} 3t = \quad + 2k\pi \\ 3t = \quad + 2k\pi \end{cases}$ $k \in \mathbb{Z} \iff \begin{cases} t = \dfrac{2\pi}{9} + k\dfrac{2\pi}{3} \\ t = \dfrac{4\pi}{9} + k\dfrac{2\pi}{3} \end{cases}$ $k \in \mathbb{Z}$

ⓑ (suite) pour les solutions sur $[0;\pi]$

$S = \left\{ \dfrac{2\pi}{9}; \dfrac{4\pi}{9}; \dfrac{8\pi}{9} \right\}$

ⓒ

$\sin\left(2t - \dfrac{\pi}{3}\right) \geq \dfrac{\sqrt{3}}{2} = \sin\left(\dfrac{\pi}{3}\right)$

On pose $X = 2t - \dfrac{\pi}{3}$;

$t \in \left[0;\dfrac{\pi}{3}\right] \iff 2t \in \left[0;\dfrac{2\pi}{3}\right] \iff \underbrace{2t - \dfrac{\pi}{3}}_{X} \in \left[-\dfrac{\pi}{3};\dfrac{\pi}{3}\right]$

$\sin X \geq \dfrac{\sqrt{3}}{2}$ pour $X = \dfrac{\pi}{3}$ sur $\left[-\dfrac{\pi}{3};\dfrac{\pi}{3}\right]$

donc $2t - \dfrac{\pi}{3} = \dfrac{\pi}{3} \iff t = \dfrac{\pi}{3}$ $\quad \boxed{S = \left\{\dfrac{\pi}{3}\right\}}$

Ex4 : $f(2\pi + x) = \sin(x + 2\pi) + \cos(2x + 4\pi) =$

$= \sin(x) + \cos(2x)$

Donc f est 2π-périodique $\quad f(x) \neq f(-x)$ et $f(x) \neq -f(-x)$

Donc f n'est ni paire ni impaire.

Orétude . sur $[0;2\pi]$

$f'(x) = \cos x - 2\sin(2x) = \cos(x) - 4\sin(x)\cos(x)$

$= \cos(x)(1 - 4\sin(x))$

$1 - 4\sin(x) = 0 \iff \sin(x) = \dfrac{1}{4} = \sin\big(\overbrace{\arcsin\left(\tfrac{1}{4}\right)}^{\varphi}\big)$

$\iff \begin{cases} x = \varphi + 2k\pi \\ x = \pi - \varphi + 2k\pi \end{cases}$ $\quad \varphi \in \left[0;\dfrac{\pi}{2}\right]$

Donc $S = \{\varphi; \pi-\varphi\}$ sur $[0;\pi]$

x	0		φ		$\frac{\pi}{2}$		$\pi-\varphi$		$\frac{3\pi}{2}$		2π
$\cos(x)$		+		+	0	−		−	0	+	
$1-4\sin(x)$		+	0	−		−	0	+		+	
$f'(x)$		+		−		+		−		+	
f	1	↗	$\frac{9}{8}$	↘	0	↗	$\frac{9}{8}$	↘	-2	↗	1

$f(x) = \sin(x) + 1 - 2\sin^2(x)$

a $\sin(\arcsin(\varphi)) = \dfrac{1}{4}$ donc

$f(0) = f(2\pi) = 1$

$f\left(\dfrac{\pi}{2}\right) = 0 \quad f\left(\dfrac{3\pi}{2}\right) = -2$

$f(\varphi) = \dfrac{1}{4} + 1 - 2\left(\dfrac{1}{4}\right)^2 = \dfrac{9}{8}$

$= f(\pi - \varphi)$